Reconstructing Subsurface Ocean Temperatures Using Deep Convolutional Neural Networks

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Abstract

The subsurface ocean contains important information in understanding climate dynamics, but direct measurements are limited due to high operational costs while satellite remote sensing provides an abundance of surface observations. We introduce a convolutional long short-term memory (ConvLSTM) neural network that takes in sea surface data including temperature, sea level height, and wind, and predicts subsurface temperature at given depths. Our ConvLSTM, after tuning hyperparameters, achieves its best performance when reconstructing sub-sea surface temperatures at different depths, and adding a binary land mask greatly increases Spearman's rank correlation coefficient by excluding non-ocean pixels. Moreover, a hybrid CNN–LSTM architecture outperforms the ConvLSTM baseline, likely because decoupling spatial feature extraction from temporal sequence modeling yields more robust pattern learning. These results demonstrate that surface observations, when combined with deep sequence models, can cost-effectively reconstruct subsurface thermal structure, opening new opportunities for global ocean and climate monitoring.

Our code is available at: `https://github.com/eb45/661-Final-Project`.

1 Introduction

Oceans serve as a buffer against rapid climate change. They absorb excessive heat trapped by greenhouse gases, and transport heat from the equator to the poles. Quantitative and accurate monitor of the oceans' internal thermal structure is foundational to ocean and climate studies.

However, obtaining observational data of the vast ocean is not a trivial task. Satellite remote sensing technology has provided large-scale high-resolution observation, but the electromagnetic waves of satellite remote sensing are only applicable for sea surface and cannot penetrate directly into the ocean to obtain internal data. Subsurface data are costly and sparse.

This motivates a data-driven approach to reconstruct subsurface temperature from surface variables. This project uses convolutional neural networks to infer subsurface temperature fields from sea surface observations. We take in the following surface variables: sea surface temperature (SST), sea surface height (SSH), zonal and meridional sea surface wind (uSSW and vSSW), and the latitude (lat) and longitude (lon) of the specific location (Vallis 2019). We expect the model to calculate the correct subsurface temperature at the given depth. In other words, this is a regression problem, that **at a given depth, we find a function f so that subsurface temperature $= f($SST, SSH, uSSW, vSSW, lat, lon$)$.** The physical basis of the model setup is in Supplementary Materials.

2 Related Works

Traditionally, comprehensive subsurface datasets rely on the floats or in situ observation, combined with reanalysis. Challenges include discontinued observing networks, reproducing inter-annual variability and trends, accounting for drift and bias due to air-sea flux, and improving performances in regions with sparse data (Storto et al. (2019)).

Recent advances in deep learning have motivated data-driven methods to this subject. For example, Tian et al (2022) explored the feed-forward neural network approach to reconstruct a high-resolution ocean subsurface salinity dataset; Guo et al. (2024) reconstructed subsurface thermohaline field based on deep learning and remote sensing data; Mao et al (2023) proposed a Dual Path Convolutional Neural Networks algorithm to reconstruct the subsurface temperature and salinity. There are also a variety of studies focusing on different regions (e.g. Wang et al. (2021), Qi et al. (2024)). All these researches offer promising accuracy and provide valuable references for ocean data analysis.

For this project, we use convolutional neural network trained on gridded surface data, to reconstruct subsurface temperature. Compared with published studies, our model requires significantly less training data and computational resources, while still performs well, making it more accessible for applications.

3 Methodology

3.1 Data

This project uses observation data combined with reanalysis data to train and test the model. These datasets are from different institutes, which requires preprocessing before training the model. All datasets used in this project are originally global in coverage, gridded over latitude and longitude coordinates, and stored in NetCDF format. See the details in table 1. Data sources are included in appendix.

Table 1: Summary of Dataset Sources and Resolutions

Variable	Source	Spatial grid (lat×lon)	Time range	frequency	Measurement
SST, uSSW, vSSW	ERA5	721×1440	1940-2024	monthly	satellite / reanalysis
SSH	MEaSUREs	960×2160	1992-2022	5-days	satellite / reanalysis
ST	ARGO	545×720	1960-2021	monthly	floats / reanalysis

For the data preprocessing, we convert the 5-day dataset to monthly average, extract the common time range 1992-2021, and interpolate all spatial grid to 360×480. The spatial resolution for this model is arbitrary, but the resolution should not exceed the original resolution. Therefore, the data of each input variable is a 3D matrix, with dimensions time $\times$ latitude $\times$ longitude $= 360 \times 360 \times 480$. The subsurface temperature (ST) originally has one more dimension (depth) than the input variables, while in this project we extract specific depth of the subsurface temperature, so that it has the same dimension and structure of the inputs.

3.2 Method

The ConvLSTM architecture that we modeled from Su et al (2022) is composed of two ConvLSTM Cells, each of kernel size 7, stride 1, and padding 3, and followed by a 2D convolution.

A ConvLSTM cell is similar to an LSTM cell, but instead convolutional kernels replace weight matrix multiplications. ConvLSTM cells still contain the cell state, hidden state, and forget, input and output gates found in LSTM cells.

The model was trained using the Adam optimizer and Mean Squared Error Loss criterion. ST data was split into a 2:1 ratio for the train test split—the first 20 years was used for training while the later 10 years was used for testing. This was done deliberately to optimize performance and prevent data leakage over different months or with non-chronological order.

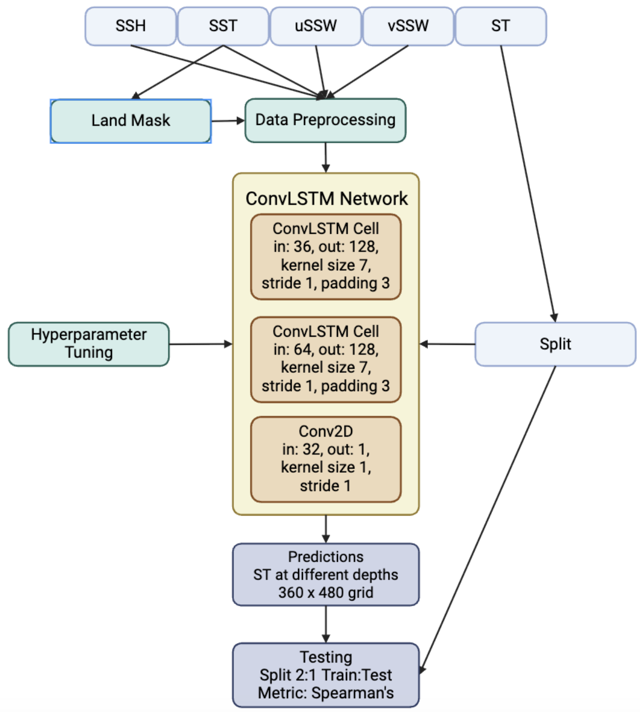


Figure 1: Model Architecture

3.3 Evaluation Metrics

We decided to utilize the Spearman's rank correlation coefficient in order to measure accuracy for our experiments.

$$\rho_s = 1 - \frac{6\sum_i d_i^2}{N(N^2-1)} \tag{1}$$

Spearman's rank correlation coefficient is particularly well suited to global, temporal oceanographic datasets that often deviate from Gaussian assumptions. By operating on the ranked values rather than their magnitudes, it quantifies whether regions exhibiting higher observed temperatures also receive correspondingly higher predictions, regardless of the underlying distribution. This rank-based criterion is inherently less sensitive to extreme anomalies—such as coastal artifacts or transient spikes—thereby yielding a more robust assessment of model performance.

4 Results

4.1 Preliminary Experiment: CNN

To test a model that does not encompass modeling both spatial and temporal data, we ran a Convolutional Neural Network (CNN) model to reconstruct subsurface ocean temperature. The results show an inability to model the temperature at different depths—the Spearman Rank Correlation coefficient was 0.6935 on testing data after training for 20 epochs, ending with a loss of 33.9650.

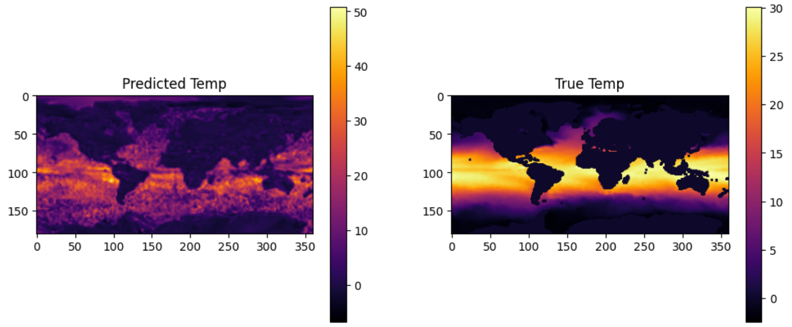
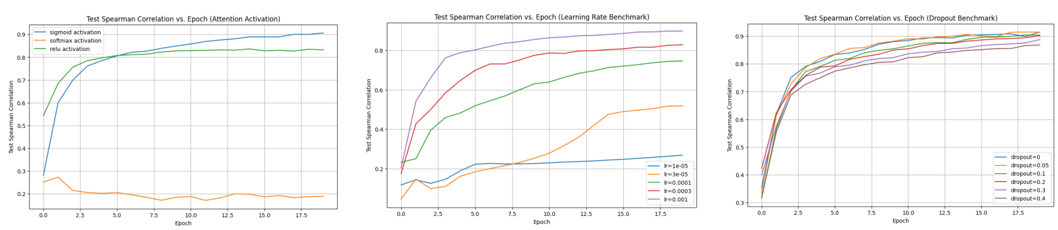


Figure 2: Predicted and True Temperature at Depth 1 using CNN

The preliminary experiment underlines the importance of using a hybrid model architecture to model both spatial and temporal data. Because subsurface ocean temperature is sensitive to both location and time, it is crucial to apply a model like the ConvLSTM to reconstruct subsurface ocean temperature.

4.2 Experiment 1: Hyperparameter Optimization

We evaluated three key hyperparameters: activation function, learning rate, and dropout—using sea-surface temperature (SST) data. Performance was measured by the Spearman rank correlation between predicted and true SST fields. Full per-configuration results are available in Supplementary Table S1.



(a) Activation functions (b) Learning rates (c) Dropout rates

Figure 3: Spearman correlation on test SST data as a function of (a) activation function, (b) learning rate, and (c) dropout rate.

Activation Function We compared Sigmoid, Softmax and ReLU variants (Figure 3a). Sigmoid achieved the highest peak correlation (0.70), outperforming Softmax and ReLU by approximately 0.05. We hypothesize that Sigmoid's smoother gradient profile mitigates transient spikes in the SST signal, leading to more stable convergence.

$$\sigma(x) = \frac{1}{1+e^{-x}}, \sigma'(x) = \sigma(x)\big(1-\sigma(x)\big) \tag{2}$$

This results in outlier damping meaning extreme inputs are mapped into $[0, 1]$, preventing large weight updates due to noisy spikes. This also results in more stable convergence as bounded updates over many time steps reduces oscillations and promotes steady descent on the loss surface.

Overall, sigmoid gating yields more robust training dynamics on noisy, temporally correlated SST sequences.

Learning Rate Learning rates in $\{1\times10^{-5}, 3\times10^{-5}, 1\times10^{-3}, 3\times10^{-3}, 1\times10^{-2}\}$ were tested (Figure 3b). A rate of 1×10^{-2} struck the best balance between rapid convergence and training stability, reaching a Spearman correlation of 0.90. Lower rates converged slowly and exhibited oscillatory behavior, while higher rates (e.g. 3×10^{-3}) showed early plateauing.

Dropout Rate We applied dropout rates from 0 to 0.4 (Figure 3c). Surprisingly, *no* dropout yielded the best performance (Spearman 0.91), suggesting that the ConvLSTM's internal gating provides sufficient regularization for SST data. Non-zero dropout slightly reduced correlation, likely by disrupting the spatio-temporal feature flow. A likely reason why zero dropout ended up yielding the highest accuracy is because we are already reducing the spatial-temporal data to a lower resolution meaning much of the data left is meaningful and thus not being overfitted.

4.3 Experiment 2: Land-Masking Ablation

We next investigate the effect of excluding land pixels from both input and loss calculations. Intuitively, land regions exhibit very different temperature dynamics than the open ocean, and including them may dilute the model's ability to learn true sea-surface patterns.

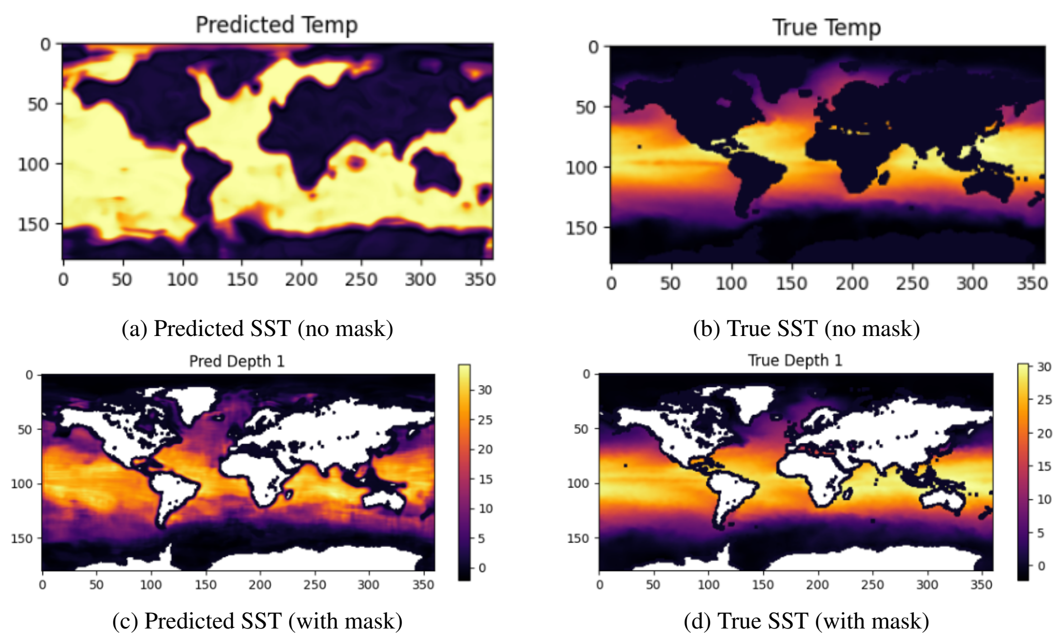
Land-Mask Definition A land-mask is a Boolean matrix $M \in \{0,1\}^{H \times W}$ where

$$M_{i,j} = \begin{cases} 1 & \text{if cell } (i,j) \text{ is ocean,} \\ 0 & \text{if cell } (i,j) \text{ is land.} \end{cases} \tag{3}$$

We compute M by first forming the multi-year mean SST at each grid cell, spatially resized to our model's 180×360 grid, and then thresholding:

$$M_{i,j} = \begin{cases} 1 & \text{if } \bar{T}_{i,j} \geq \tau, \\ 0 & \text{if } \bar{T}_{i,j} < \tau. \end{cases} \tag{4}$$

where we sweep $\tau \in \{10^{-3}, 1, 100\}$. During training, masked-out (land) cells contribute neither to the model input nor to the loss.



(a) Predicted SST (no mask)

(b) True SST (no mask)

(c) Predicted SST (with mask)

(d) True SST (with mask)

Figure 4: Comparison of sea-surface temperature fields. Top row: no land-mask applied. Bottom row: land-mask ($\tau = 10^{-3}$) applied during training.

Without land-masking (panels a–b): The model is forced to produce an SST value at every grid cell, including those over continents where no "true" ocean temperature exists. As a result, its outputs bleed across coastlines creating blurred continental outlines. This is noticeably inefficient as noted by the detrimental impact on accuracy.

With land-masking (panels c–d): By zeroing out land pixels during training, the network learns exclusively from ocean-valid data. In panel (c), coastlines are crisp, and spatial patterns emerge far more clearly.

Results Table 2 reports test Spearman correlation over three independent trials. Applying land-masking at even a loose threshold ($\tau = 1$) yields a substantial gain over the no-mask baseline, and the optimal threshold $\tau = 10^{-3}$ achieves the highest average accuracy.

Table 2: Land-Masking Ablation: Test Spearman Correlation

	No Mask	**Mask** ($\tau = 10^{-3}$)	**Mask** ($\tau = 1$)	**Mask** ($\tau = 100$)
Trial 1	0.8790	0.9267	0.9293	0.9300
Trial 2	0.8706	0.9297	0.9225	0.9243
Trial 3	0.8715	0.9185	0.9170	0.9187
Avg	0.8737	**0.9250**	0.9229	0.9243

4.4 Experiment 3: Hybrid Architecture Comparison

For the final experiment, we trained a CNN-LSTM model, a different hybrid architecture that, like the ConvLSTM, also utilizes both 2D convolutions and LSTMs. However, while ConvLSTMS perform convolution operations on the input, forget, and output gate information in the hidden state of LSTM cells and memory cells from the previous timestep and the current timestep, convolutions in CNN-LSTMS are performed outside the LSTM recurrent cell. Instead, CNN-LSTMs follow an architecture of 2D convolutions and max pooling, followed by the LSTM and a fully-connected layer.

Reconstructing subsurface ocean temptures with the CNN-LSTM model resulted in a Spearman coefficient ρ of 0.9689. A visual of the difference between our train and predicted temperatures at varying depths corroborates this (Figure S2). Our results indicate that even without hyperparameter benchmarking and testing different microarchitectures, CNN-LSTMs performed better than ConvLSTMs. This performance suggests that modeling spatial data and then temporal data yield better reconstruction of subsurface ocean temperature than modeling both at the same time. By first learning the spatial data from filters in the convolution first and then running the LSTM on temporal data from the compact feature vector, the model appears to result in a higher accuracy. The LSTM is thus able to focus on temporal dependencies without looking at spatial features at the same time. This may also be attributed to the CNN-LSTM having a larger spatial receptive field, so broad data patterns like SST and SSH is better captured.

Although recent papers discussing modeling subsurface ocean temperatures mostly use ConvLSTMs, our results show that CNN-LSTMs may prove to be useful in this task.

5 Conclusion

This paper aims to demonstrate that subsurface ocean temperature can be successfully reconstructed using sea surface data. We trained a Convolution Long-Short Term Memory model (ConvLSTM) composed of two layers of ConvLSTM cells, followed by a 2D convolution, no dropout, and used the sigmoid attention activation function with a learning rate of 0.01 run on 20 epochs. After iterative rounds of testing, these hyperparameters proved to produce the most accurate results. While the model performs better in the near surface and deep ocean, it shows reduced accuracy in regions with high spatial and temporal variability (see Figure **??** below).

Introducing a binary land mask at a 1×10^{-3} threshold boosted predictive Spearman correlation by roughly 5%, confirming that excluding non-ocean data significantly enhances the model's ability to learn the true subsurface ocean temperature patterns.

The CNN-LSTM achieved a Spearman coefficient of 0.9689, outperforming the ConvLSTM originally outlined in our reference paper, likely because decoupling spatial feature extraction from temporal sequence modeling enables more robust learning of patterns. The comparison of both hybrid models to just CNN highlights the significance of hybrid approaches that utilize both spatial and temporal modeling.

We recognize there may be some limitations in our approach: performance is limited by the coverage of training data and sensitive to the quality of surface input data like sea surface temperature, height, and wind; any noise or errors in surface data may propagate into the subsurface estimates. Because our model is purely data-driven, it lacks explicit physical constraints. As a result, a future study could incorporate physical constraints into the data-driven model to enhance the prediction accuracy, which could help improve both accuracy and efficiency by guiding the network with fundamental ocean dynamics, while retaining the adaptability and data-driven strengths of deep learning.

Acknowledgements

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Supplementary Material

A Introduction

Physical Link between Surface and Subsurface Is it reasonable to reconstruct subsurface ocean temperature from surface data? Why do we use SSH, SST, and SSW, rather than other surface variables? According to fluid mechanics, the thermodynamics equation of the ocean is

$$\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T = \nabla \cdot (\kappa \nabla T) + Q \tag{5}$$

where T is the temperature of the ocean, which is of interest in this project. κ is the diffusion constant; Q is the external forcing; $\mathbf{u}$ is the 3D velocity field, governed by the momentum equations

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} + f\mathbf{k} \times \mathbf{u} = -\frac{\nabla p}{\rho} + \nu \nabla^2 \mathbf{u} + \mathbf{F} \tag{6}$$

where f is the Coriolis coefficient, ρ is the water density, ν is viscosity coefficient, p is the pressure, and $\mathbf{F}$ is the wind forcing. With the continuum equation $\nabla \cdot \mathbf{u} = 0$, the whole ocean velocity and temperature structure are determined.

From the equations above, we shall identify the external terms depending on surface variables: external heat Q, pressure p, and wind forcing $\mathbf{F}$. Other terms are determined by inherent properties of water, fluid, or the topography and location. This is the theoretical basis of this data-driven model of this project.

Recent years, with the development of deep ocean remote sensing and the float-based observation, subsurface datasets become available. These datasets are not as high-resolution as surface datasets, but provide resources to train the model.

The surface variables taken into the model are: sea surface temperature (SST), sea surface height (SSH), zonal and meridional sea surface wind (uSSW and vSSW), and the latitude and longitude of the specific location. Here, SST reflects the external heat Q, SSH reflects the boundary condition of the pressure p, and the surface wind reflects the $\mathbf{F}$ term. Therefore, in terms of the physical significance, the model is trained to learn the non-linear interaction within the equations.

B Data sources

SST, uSSW, vSSW: ERA5 monthly averaged data on single levels from 1940 to present Hersbach, H., Bell, B., Berrisford, P., Biavati, G., Horányi, A., Muñoz Sabater, J., Nicolas, J., Peubey, C., Radu, R., Rozum, I., Schepers, D., Simmons, A., Soci, C., Dee, D., Thépaut, J-N. (2023): ERA5 monthly averaged data on single levels from 1940 to present. Copernicus Climate Change Service (C3S) Climate Data Store (CDS), DOI: 10.24381/cds.f17050d7

SSH: MEaSUREs Gridded Sea Surface Height Anomalies Version 2205 Fournier S., Willis J., Killett E., Qu Z. and Zlotnicki V.. 2022. JPL MEaSUREs Gridded Sea Surface Height Anomalies Version 2205. Ver. 2205. PO.DAAC, CA, USA. https://doi.org/10.5067/SLREF-CDRV3

ST: Coriolis Ocean Dataset for Reanalysis Cabanes, C., A. Grouazel, K. von Schuckmann, M. Hamon, V. Turpin, C. Coatanoan, F. Paris, S. Guinehut, C. Boone, N. Ferry, C. de Boyer Montégut, T. Carval, G. Reverdin, S. Pouliquen, and P. Y. Le Traon, 2013: The CORA dataset: validation and diagnostics of in-situ ocean temperature and salinity measurements. Ocean Science, 9, 1-18, http://www.ocean-sci.net/9/1/2013/os-9-1-2013.html, doi:10.5194/os-9-1-2013.

Sample surface data visualized:

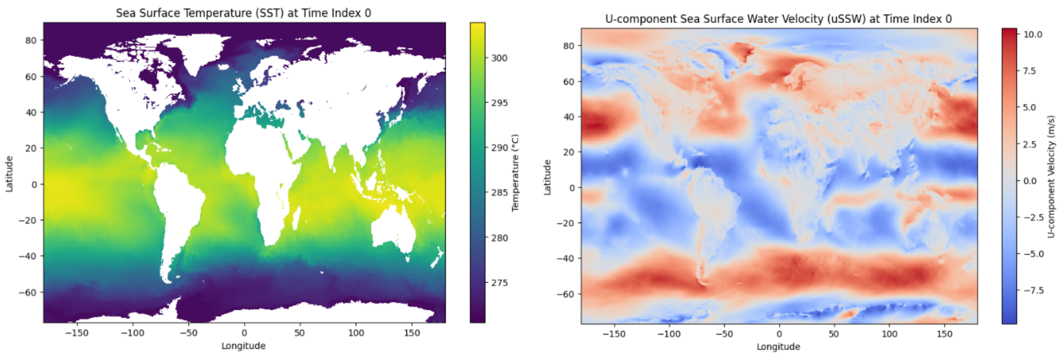

Figure S1: Sample of input data. Sea surface temperature on the left, zonal sea surface wind on the right.

C Experiment I

Table S1: Best Test Spearman ρ per Hyperparameter Setting

Activation	LR	Dropout	Epoch	Train Loss	Spearman
Activation Function (lr=0.001, dropout=0.2)					
sigmoid	0.001	0.2	20	17.9897	0.9063
softmax	0.001	0.2	2	174.1949	0.2735
relu	0.001	0.2	15	27.1613	0.8360
Learning Rate (sigmoid, dropout=0.2)					
sigmoid	1e-05	0.2	20	117.9336	0.2684
sigmoid	3e-05	0.2	20	100.5948	0.5182
sigmoid	0.0001	0.2	20	62.1776	0.7460
sigmoid	0.0003	0.2	20	34.6419	0.8284
sigmoid	0.001	0.2	20	17.9897	0.9063
Dropout (sigmoid, lr=0.001)					
sigmoid	0.001	0	20	11.4244	0.9146
sigmoid	0.001	0.05	19	13.9991	0.9144
sigmoid	0.001	0.10	19	16.4435	0.9038
sigmoid	0.001	0.20	20	17.9897	0.9063
sigmoid	0.001	0.30	20	22.3322	0.8873
sigmoid	0.001	0.40	20	28.2451	0.8679

D Experiment III

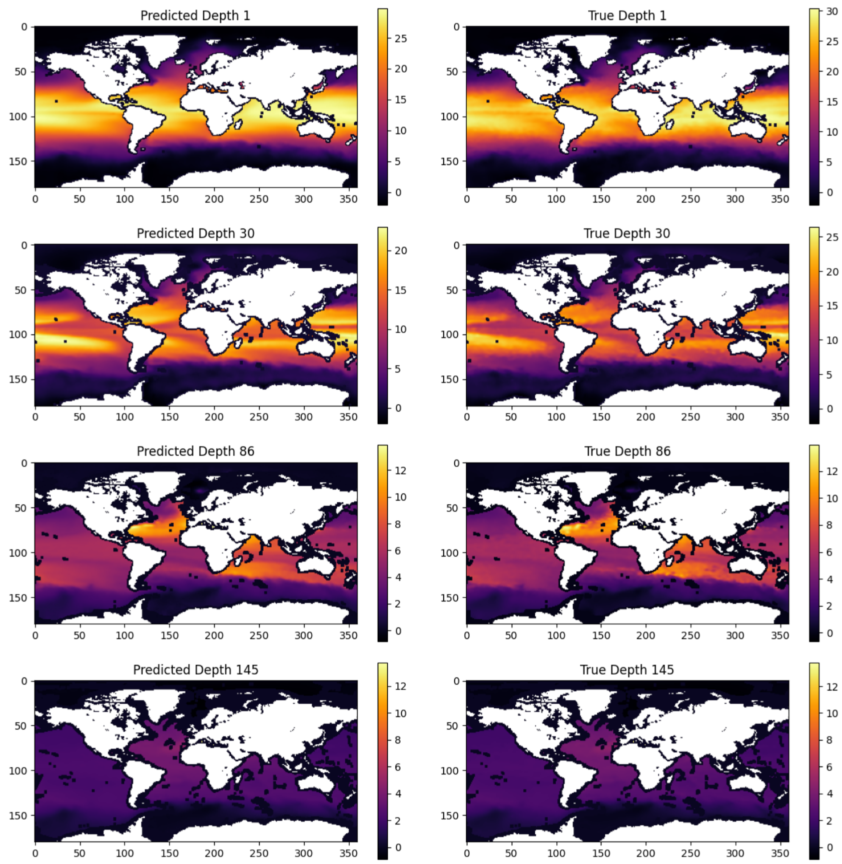

Figure S2: Predicted and True Temperatures at Varying Depths

E Discussion

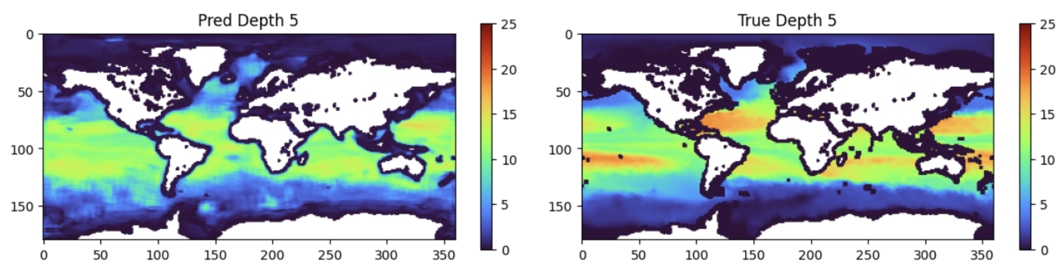

Figure S3: A sample showing the difference between reconstructed and true subsurface temperature. In this sample, the model fails to capture the characteristics over mid-latitude.